

Deep Learning for Tornadoic Supercell Detection

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Abstract

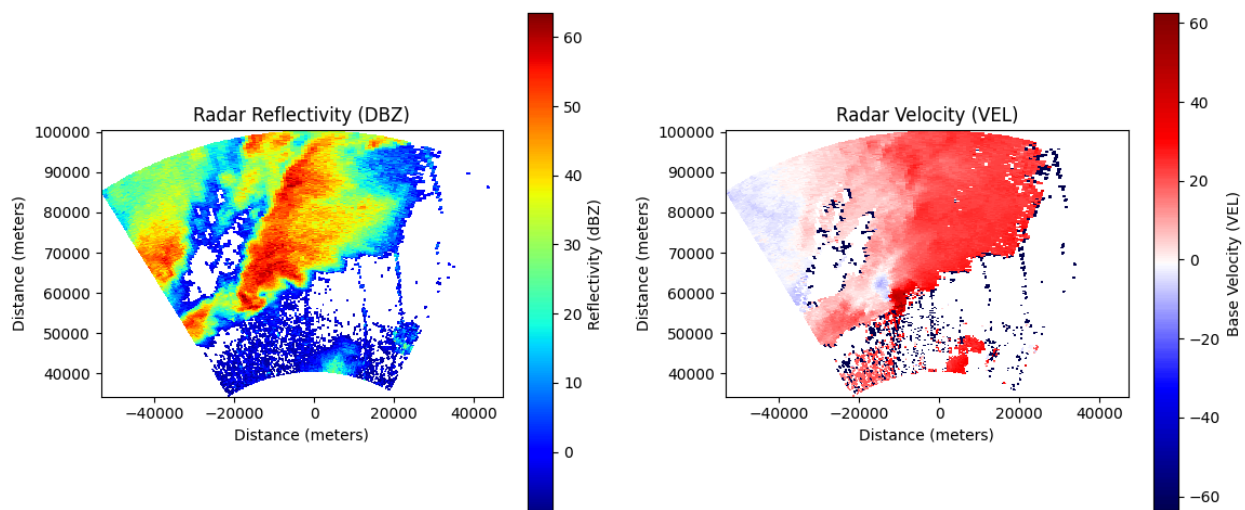
This project aims to improve tornado detection from Doppler radar data using deep learning. Existing systems in use by the National Weather Service (NWS) rely on basic machine learning models and hard-coded algorithms, both of which require a knowledgeable human in the loop, to issue their tornado warnings and emergencies. This system often produces false negatives (NOAA, 2024), and has in the past failed to issue timely warnings, resulting in excess deaths and injuries. Leveraging the TorNet dataset, which contains radar images of storms for which the NWS issued warnings between 2013 and 2022 (Veillette, Kurdzo, et. al), we trained a convolutional neural network (CNN) to classify storms as tornadoic or non-tornadoic. Despite challenges such as dataset size, class imbalance, and noisy data, our final model achieved a test accuracy of 99%, precision of 0.98, and recall of 0.97. This represents a significant improvement over initial models and NWS precision. Limitations include difficulty detecting non-supercell tornadoes and the absence of temporal analysis in the model. Future work includes exploring recurrent neural networks (RNNs) to better capture temporal data, as well as incorporating additional data such as current local and national weather data and local storm trends. This report details our methods, experiments, results, and the implications of our findings for enhancing tornado warning systems.

Introduction

Tornadoes pose a direct and severe threat to life and property. Early detection is crucial for mitigating fatalities, as a six-minute improvement in warning times for tornadoes can reduce expected casualties by roughly 40% (Simmons, Sutter et. al., 2008). Approximately 80% of tornadoes (including 90-95% of severe tornadoes), are spawned by storms known as supercells, which feature a strong updraft (like a traditional thunderstorm), but with the caveat that the updraft is rotating. This feature results in the

consistent and clear appearance of these storms in radar data- this appearance is known as a Tornado Vortex Signature (TVS), and features a “hook echo” (a hook-shaped region of precipitation where the rotating winds wrap rain around the back of the storm, next to a region of low or no precipitation, marking the updraft of the storm) as well as a “velocity couplet” (a tight couplet of wind moving towards the radar unit, next to wind moving away from the radar unit, representing rotation below the cloud base) (Murdoch, 2016).

Both traditional methods and our approach take advantage of TVSSs. Current NWS systems employ basic machine learning to flag these features, but suffer from frequent false alarms and missed detections (NOAA, 2024). Our project addresses these shortcomings by using CNNs to directly analyze radar imagery for tornadic signatures. We hypothesize that deep learning can offer an improvement in precision without sacrificing recall, reducing the need for human involvement in the decision loop, allowing warnings to be issued more quickly. We used the TorNet dataset for this project, which contains Doppler radar data from all storms that received any type of severe weather warning by the NWS between 2013 and 2022. This dataset also contained labels (tornadic or non-tornadic) in a separate catalog file, which served as the target for our training data.



Background

Prior work in this field has primarily focused on the use of deep learning to process atmospheric variables, in order to make relative risk predictions for certain types of severe weather over large areas (on the scale of multiple states) and over large time frames (days/weeks) (Hempel, 2024). One such example is NadoCast, a model that predicts multi-day tornado probabilities, along with wind and hail risks, using atmospheric soundings (data collected by weather balloons) (Murdoch, 2016). Additionally, deep learning approaches have been used on doppler radar data to predict rain, snowfall, and wind patterns. Current approaches preprocess radar data into heavily engineered derivatives before analysis, while our model utilizes the raw radar products mentioned above (DBZ and VEL) to maximize data fidelity and keep fine-grained information critical for TVS identification.

Methods

Radar units are capable of detecting the presence of precipitation (and other objects) in the atmosphere- hence their application in weather forecasting. This radar product is known as “base reflectivity”, often denoted as “DBZ”. This is the radar product most of us are intimately familiar with, from seeing radar loops on weather apps or on TV. Additionally, Doppler radar units are also capable of detecting the relative direction of movement of this precipitation, via the Doppler effect, hence the name of these radar units. This radar product displays whether objects in a certain region are moving relatively towards or away from the radar unit, and is known as “base velocity”, or “VEL”. While there exist several more radar products useful for general purpose weather forecasting, these two are the most useful for tornado detection. There exists a network of Doppler radar units whose field of view collectively covers the entire continental United States, allowing us to view these data points for any given storm.

Experiments

The TorNet dataset contains Doppler radar data from severe storms in the U.S. spanning 2013-2022. Each entry includes four radar images taken during the first 20 minutes after the storm’s appearance on radar, one from each sweep of the radar beam. The data’s size (150GB compressed), and

class imbalance (80% non-tornadic) presented significant challenges. Importing the data all at once was impossible, as uncompressing the data would have used more RAM than was available on even Google Colab Pro's high-RAM GPU's. Instead, we chose to uncompress one year's worth of data at a time, extract the features we needed, process the data fully, and delete the original files and intermediate data before proceeding with the next year's data.

We slimmed the dataset to relevant radar products (DBZ and VEL) and labels, replaced NaN values with 0 (most radar images are made up of NaN values, as most of the area visible to the radar unit at a given time will not contain any precipitation), and finally normalized and scaled values and resized the radar images (which were initially in polar coordinates, due to the inherent polar nature of radar data) to 256x256, preparing them for use in a CNN. Each radar sweep contained two beam angles. Initially, we believed this would be useful to our model as this would allow it to discern between true TVS's and incidental storm features, but we did not see a meaningful difference in models trained on only the standard radar beam angle versus those trained on both angles, so we chose to remove the higher angle from our dataset altogether and thus reduce the dimensionality of our data while cutting its size roughly in half. Our final processed dataset contained approximately 70GB of data.

Our initial model was built to test whether this project was a realistic application of deep learning, and how deep learning would perform on this dataset. We initially built a fully-connected neural network with two hidden layers, with a binary classification as output (tornadic vs non-tornadic), training it on one year's worth of data, and achieving a test accuracy of 82%. Because our dataset contained radar data from all severe storm warnings issued in the U.S. between 2013 and 2022, we naturally had a major class imbalance- 80% of the storms in the dataset were non-tornadic. A model could achieve 80% testing accuracy by simply identifying all testing instances as non-tornadic. Therefore, testing accuracy alone is not a satisfactory benchmark for our models. We chose to consider precision and recall to determine the efficacy of our model, as our main goal in taking on this project was to improve on the NWS's precision value of approximately 0.5 (Murdoch, 2016). In this case, precision was 0.77, and recall was 0.78, a

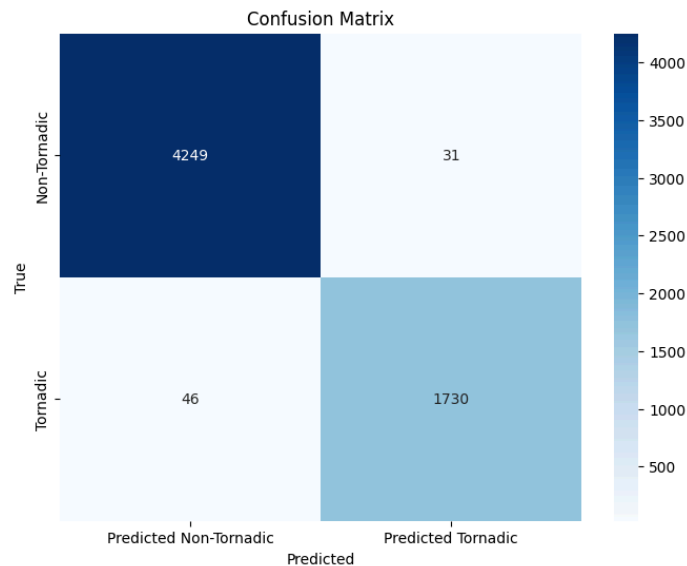
promising start, but with much room for improvement. We also noted that this model began overfitting less than 10 epochs into training, and incorporated dropout layers into our final model to combat this.

Our final model, we implemented a CNN. The model architecture had the following layers: three convolutional blocks with Batch Normalization and MaxPooling, which were used to prevent overfitting and speed up training; two dense layers for classification, including dropout layers to further combat overfitting; and our output, a binary classifier. We used Google Colab Pro with A100 GPUs, training over 20 epochs with class weights to address the class imbalance. Training this model on the entire training dataset took approximately seven hours.

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 118, 238, 32)	608
max_pooling2d (MaxPooling2D)	(None, 59, 119, 32)	0
batch_normalization (BatchNormalization)	(None, 59, 119, 32)	128
conv2d_1 (Conv2D)	(None, 57, 117, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 28, 58, 64)	0
batch_normalization_1 (BatchNormalization)	(None, 28, 58, 64)	256
conv2d_2 (Conv2D)	(None, 26, 56, 128)	73,856
max_pooling2d_2 (MaxPooling2D)	(None, 13, 28, 128)	0
batch_normalization_2 (BatchNormalization)	(None, 13, 28, 128)	512
flatten (Flatten)	(None, 46592)	0
dense (Dense)	(None, 128)	5,963,904
batch_normalization_3 (BatchNormalization)	(None, 128)	512
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 64)	8,256
batch_normalization_4 (BatchNormalization)	(None, 64)	256
dropout_1 (Dropout)	(None, 64)	0
dense_2 (Dense)	(None, 1)	65

The transition to a CNN improved results astoundingly. The test accuracy improved to 99%, the precision to 0.98, and the recall to 0.97. The precision of our model significantly outperforms the NWS's ~50% false positive rate. While the NWS has historically failed to issue warnings for dangerous storms,

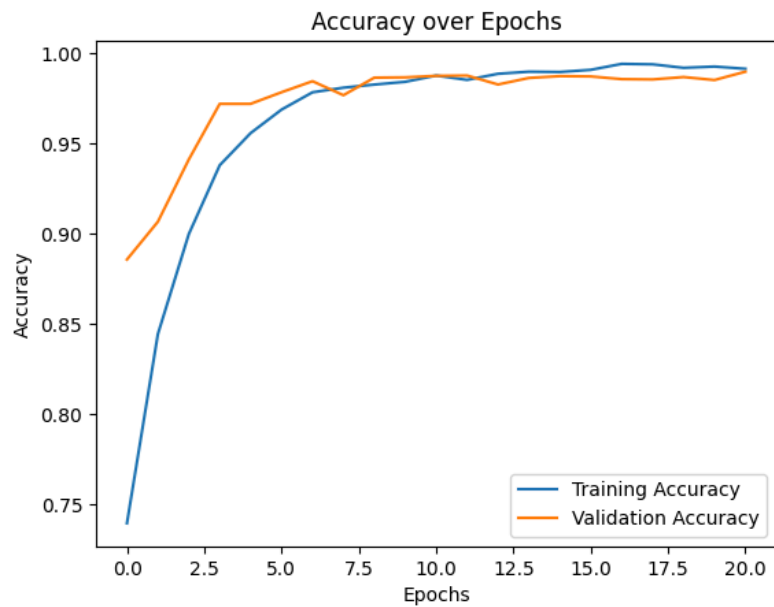
these events are a rarity and the NWS's recall is effectively 1.0 (Murdoch, 2016), a hard number to compete with, though our model's recall does come very close. While we could sacrifice some of our high precision to increase our recall to be as close to that of the NWS as possible, we have chosen to stick with our current classification threshold, as there is an important reason for the few false negatives our model



outputs.

This model excels in detecting tornadic supercells but faces limitations in detecting non-supercell tornadoes. These storms lack clear TVS signatures, as the storm features common to supercells are not present, and the radar data looks more like a standard thunderstorm. Around 20% of tornadoes come from non-supercell storms, so these types of tornadoes made up approximately 4% of our dataset, limiting how often the model sees examples of this type of storm in training. This limits the recall on our final model, and introduces a theoretical soft ceiling on the recall at around 0.96, after which the model would have to correctly identify these storms to improve recall- our final recall value was 0.97, suggesting that our model is potentially correctly identifying some of these instances, despite their rarity and less clear data.

One final result of note that is important to understand is the high initial training and validation accuracy of our model. Again, this is likely due to the class imbalance in our data- 80% non-tornadic.



Discussion

An obvious next step for future work would be the development of an RNN. Our data contains images of storms across four time intervals each, and this could be used to identify differences in early storm development between storms that will eventually produce tornadoes and those that will not. Additionally, the second sweep angle could be used for slight improvements to the current model, as it would identify rotation higher in the storm, signifying a strong rotating updraft. Another aspect that could be incorporated is additional data- DBZ and VEL are not the only radar products that Doppler radar units are capable of observing. For example, “correlation coefficient” (RHOHV) tracks how uniform the size of precipitation (or other objects) in the atmosphere are in a given region. This radar product is highly useful to the detection of hail, but it can also detect “debris balls” which occur when powerful tornadoes lift debris high into the atmosphere (Murdoch, 2016). As a logical extension, inspiration could be taken from existing models that operate on derivatives of radar data- additional processing could be done on the data to extract the TVS more clearly. Finally, an aspect we ignored in this project, which is nevertheless highly important to the development of tornadoes from a given storm, is atmospheric data. Using local

atmospheric soundings, measuring pressure, humidity, temperature, and wind at different altitudes, may be useful in determining whether an existing supercell may turn tornadic.

Conclusion

Our CNN model demonstrates that deep learning can enhance tornado detection, reducing false alarms and potentially improving response times. The testing accuracy (99%), precision (0.98) and recall (0.97) of our final model are competitive with the current NWS model, and there are clear paths to additional improvements through the use of additional radar products and derivatives, additional data processing, and the introduction of atmospheric data, as well as the development of an RNN to build on the success of the current CNN model.

References

Cuomo, Joaquin, and V. Chandrasekar. "Use of Deep Learning for Weather Radar Nowcasting."

AMETSOC, American Meteorological Society, 1 Sept. 2021,
journals.ametsoc.org/view/journals/atot/38/9/JTECH-D-21-0012.1.xml.

Foy, Kylie. "An AI Dataset Carves New Paths to Tornado Detection." *MIT Lincoln Laboratory*,

www.ll.mit.edu/news/ai-dataset-carves-new-paths-tornado-detection.

Hempel, Brian. "Brianhempel/NADOCAST: Tornado Probabilities via Post-Processing Weather Model

Outputs with Machine Learning." *GitHub*, github.com/brianhempel/nadocast?tab=readme-ov-file.

Murdoch, Gregory. "A Case Study in Tornado Forecasting." *National Weather Service*, NOAA's National

Weather Service, 26 Sept. 2016, www.weather.gov/maf/research_torforecast.

Simmons, Kevin M., and Daniel Sutter. "Tornado Warnings, Lead Times, and Tornado Casualties: An Empirical Investigation." *AMETSOC*, American Meteorological Society, 1 Apr. 2008, journals.ametsoc.org/view/journals/wefo/23/2/2007waf2006027_1.xml.

"Tornadoes." *NOAA National Severe Storms Laboratory*, www.nssl.noaa.gov/research/tornadoes/.

US Department of Commerce, NOAA. "Tornado Safety." *National Weather Service*, NOAA's National Weather Service, 9 July 2024, www.weather.gov/safety/tornado.

Veillette, Mark, and James Kurdzo. "A Benchmark Dataset for Tornado Detection and Prediction Using Full-Resolution Polarimetric Weather Radar Data." *A Benchmark Dataset for Tornado Detection and Prediction Using Full-Resolution Polarimetric Weather Radar Data*, arxiv.org/html/2401.16437v1.